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import csv
import sqlite3
from collections import Counter
from pathlib import Path

DB_PATH = Path("shipment_database.db")
DATA_DIR = Path("data")

SHEET_0 = DATA_DIR / "shipping_data_0.csv"
SHEET_1 = DATA_DIR / "shipping_data_1.csv"
SHEET_2 = DATA_DIR / "shipping_data_2.csv"

def get_or_create_product_id(cursor, product_name):
    """Return the id of `product_name` in the product table, inserting it
    if it doesn't already exist."""
    cursor.execute("SELECT id FROM product WHERE name = ?", (product_name,))
    row = cursor.fetchone()
    if row is not None:
        return row[0]

    cursor.execute("INSERT INTO product (name) VALUES (?)", (product_name,))
    return cursor.lastrowid

def insert_shipment(cursor, product_name, quantity, origin, destination):
    product_id = get_or_create_product_id(cursor, product_name)
    cursor.execute(
        """
        INSERT INTO shipment (product_id, quantity, origin, destination)
        VALUES (?, ?, ?, ?)
        """,
        (product_id, quantity, origin, destination),
    )

def process_sheet_0(cursor):
    """Spreadsheet 0 is self-contained: one product/shipment per row."""
    with open(SHEET_0, newline="") as f:
        reader = csv.DictReader(f)
        for row in reader:
            insert_shipment(
                cursor,
                product_name=row["product"],
                quantity=int(row["product_quantity"]),
                origin=row["origin_warehouse"],
                destination=row["destination_store"],
            )

def process_sheets_1_and_2(cursor):
    """Spreadsheet 1 has one row per product unit per shipment; spreadsheet
    2 maps each shipment_identifier to its origin/destination. We group
    spreadsheet 1 by (shipment_identifier, product) to get quantities, then
    join with spreadsheet 2 for origin/destination."""

    # Build a lookup of shipment_identifier -> (origin, destination)
    shipment_locations = {}
    with open(SHEET_2, newline="") as f:
        reader = csv.DictReader(f)

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        for row in reader:
            shipment_locations[row["shipment_identifier"]] = (
                row["origin_warehouse"],
                row["destination_store"],
            )

# Count how many units of each product appear per shipment
product_counts = Counter()
with open(SHEET_1, newline="") as f:
    reader = csv.DictReader(f)
    for row in reader:
        key = (row["shipment_identifier"], row["product"])
        product_counts[key] += 1

# Insert one row per (shipment, product) group, with the summed quantity
for (shipment_id, product_name), quantity in product_counts.items():
    origin, destination = shipment_locations[shipment_id]
    insert_shipment(cursor, product_name, quantity, origin, destination)

def main():
    conn = sqlite3.connect(DB_PATH)
    cursor = conn.cursor()

    process_sheet_0(cursor)
    process_sheets_1_and_2(cursor)

    conn.commit()
    conn.close()
    print("Database populated successfully.")

if __name__ == "__main__":
    main()

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